

# SECURITY ASSESSMENT REPORT

Automated Incident Response & Vulnerability Scan Analysis

Target IP Address: 8.8.8.8

Scan Timestamp: 2026-08-26T18:26:58Z

Report Generated: 2026-08-26 18:27:13 UTC

OVERALL THREAT RISK LEVEL

CRITICAL

# 1. Executive Summary & AI Analysis

# Security Assessment – 8.8.8.8 (Google Public DNS)

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## Risk Overview

### Overall Rating: LOW

The scan only reveals the expected public DNS (TCP 53) and HTTPS (TCP 443) services that Google deliberately exposes. All listed CVEs are legacy vulnerabilities (1999-2000) that do **not** apply to modern, patched Google services. Consequently, the immediate exploitable risk from the data provided is minimal.

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## Prioritised Vulnerabilities

| Rank | CVE / Finding | Why it's (not) dangerous | Exploitability |

|-----|-----|-----|-----|

| 1 | **All listed CVEs** (CVE-1999-1115, CVE-1999-1493, CVE-1999-0521, CVE-1999-1402, CVE-1999-0505, CVE-1999-1537, CVE-2000-0539, CVE-2000-0739, CVE-2000-0740, CVE-2000-0791) | These vulnerabilities target obsolete operating systems (HP Apollo Domain/OS, early Windows NT, IIS 3/4, JRun 2.3, Solaris 2.x, etc.). Google's DNS and front-end infrastructure runs on modern, fully patched Linux/Google custom stacks; none of the vulnerable software versions are present. | **N/A** – not exploitable on this host. |

| 2 | **Open TCP 53 (DNS)** | Legitimate public DNS service; exposure is intentional. | Low – requires a DNS-specific exploit (e.g., cache poisoning). No known unpatched DNS-related CVEs were reported in the scan. |

| 3 | **Open TCP 443 (HTTPS)** | Public HTTPS endpoint (likely a load balancer / API). | Low – would need a vulnerability in the web stack; none were identified. |

*Conclusion:* No actionable high-severity vulnerabilities are present in the data set.

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## Critical Attack Vectors

| Port | MITRE ATT&CK; Technique | Likely attacker path |

|-----|-----|-----|

| 53 | **T1046 – Network Service Discovery** | Reconnaissance: an adversary probes the IP to confirm DNS service is reachable. |

| 443 | **T1190 – Exploit Public-Facing Application** | If a zero-day or unknown flaw existed in the HTTPS service, an attacker could attempt remote code execution. No such flaw is reported. |

Because the services are public and heavily hardened, the most realistic vector is **reconnaissance** (T1046) followed by **credential-only attacks** (e.g., abusing DNS query rate-limiting) rather than direct exploitation of the listed CVEs.

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## Immediate Remediation Steps

> **Note:** As the target is a third-party public service (Google), you cannot directly remediate it. The steps below are for any **internal** DNS/HTTPS services that might resemble the scan results.

- **Validate Service Versions**

```
`bash
```

```
# Example for a Linux DNS server (BIND)
```

```
named -V
```

```
# Example for HTTPS (Apache/Nginx)
```

```
apache2 -v # or nginx -v
```

```
`
```

Confirm that the software is a supported, up-to-date release.

- **Apply Security Patches**

- For BIND: `apt-get update && apt-get install bind9` (or use your distro's package manager).
- For web servers: update to the latest stable release; enable automatic security updates.

- **Restrict Unnecessary Protocols**

- If TCP 53 is only needed for **UDP** queries, block TCP on the firewall:

```
`bash
```

```
iptables -A INPUT -p tcp --dport 53 -j DROP
```

```
`
```

- Close any unused ports (e.g., 443 if the service is not required).

- **Enforce Strong TLS**

- Disable weak ciphers and TLS 1.0/1.1. Example for Nginx:

```
`nginx
```

```
ssl_protocols TLSv1.2 TLSv1.3;
```

ssl\_ciphers HIGH:!aNULL:!MD5;

,

- **Enable Logging & Monitoring**

- DNS: enable query logging and forward logs to a SIEM.
- HTTPS: enable access/error logs, monitor for anomalous request patterns.

- **Deploy IDS/IPS**

- Deploy a network-based IDS (e.g., Suricata, Zeek) to alert on known exploit signatures for DNS or HTTPS.

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## Long-Term Hardening Recommendations

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| Area | Recommendation |

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| **Patch Management** | Adopt a formal, automated patch cycle (e.g., OS + application + firmware). Verify patch compliance weekly. |

| **Network Segmentation** | Isolate DNS servers in a dedicated VLAN with strict inbound/outbound ACLs; only allow trusted resolvers. |

| **Zero-Trust Access** | Use mutual TLS or client certificates for any administrative access to DNS/HTTPS services. |

| **Secure Configuration Baselines** | Adopt CIS Benchmarks for BIND, Apache/Nginx, and OS hardening. Automate compliance checks with tools like OpenSCAP or Chef InSpec. |

| **Rate-Limiting & DoS Protection** | Enable DNS query rate limiting (`rate-limit { responses-per-second 5; }`) and configure web application firewalls (WAF) for HTTPS endpoints. |

| **Threat Intelligence Integration** | Subscribe to CVE feeds and automatically map new findings to your asset inventory; prioritize only those affecting current versions. |

| **Incident Response Playbooks** | Develop specific playbooks for DNS compromise and web application exploitation (contain, eradicate, recover). Conduct tabletop exercises quarterly. |

| **Regular Penetration Testing** | Schedule external and internal pen tests at least annually to validate that no legacy or misconfigured services are exposed. |

| **Logging Retention & Analysis** | Retain DNS and HTTPS logs for  $\geq 90$  days; implement automated anomaly detection (e.g., using Elastic Stack or Splunk). |

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Bottom Line

The scan results do not reveal any exploitable, current day vulnerabilities on the public IP 8.8.8.8. The listed CVEs are historical and irrelevant to Google's infrastructure. Focus your effort on **hardening any internal DNS/HTTPS services** you control, ensuring they run up to date software, are properly firewalled, and are continuously monitored.

2. Open Ports & Detected Services

| Port    | Service | Version | Risk   |
|---------|---------|---------|--------|
| 53/tcp  | domain  | N/A     | LOW    |
| 443/tcp | https   | N/A     | MEDIUM |

3. Discovered Vulnerabilities (CVEs)

CVE-1999-1115 — CVSS Score: 7.2 | Published: N/A

HIGH SEVERITY

Vulnerability in the /etc/suid\_exec program in HP Apollo Domain/OS sr10.2 and sr10.3 beta, related to the Korn Shell (ksh).

CVE-1999-1493 — CVSS Score: 10.0 | Published: N/A

CRITICAL SEVERITY

Vulnerability in crp in Hewlett Packard Apollo Domain OS SR10 through SR10.3 allows remote attackers to gain root privileges via insecure system calls, (1) pad\_\$dm\_cmd and (2) pad\_\$def\_pfk().

CVE-1999-0521 — CVSS Score: 7.2 | Published: N/A

HIGH SEVERITY

An NIS domain name is easily guessable.

CVE-1999-1402 — CVSS Score: 2.1 | Published: N/A

LOW SEVERITY

The access permissions for a UNIX domain socket are ignored in Solaris 2.x and SunOS 4.x, and other BSD-based operating systems before 4.4, which could allow local users to connect to the socket and possibly disrupt or control the operations of the program using that socket.

**CVE-1999-0505 — CVSS Score: 7.2 | Published: N/A****HIGH SEVERITY**

A Windows NT domain user or administrator account has a guessable password.

**CVE-1999-1537 — CVSS Score: 5.0 | Published: N/A****MEDIUM SEVERITY**

IIS 3.x and 4.x does not distinguish between pages requiring encryption and those that do not, which allows remote attackers to cause a denial of service (resource exhaustion) via SSL requests to the HTTPS port for normally unencrypted files, which will cause IIS to perform extra work to send the files over SSL.

**CVE-2000-0539 — CVSS Score: 6.4 | Published: N/A****MEDIUM SEVERITY**

Servlet examples in Allaire JRun 2.3.x allow remote attackers to obtain sensitive information, e.g. listing HttpSession ID's via the SessionServlet servlet.

**CVE-2000-0739 — CVSS Score: 5.0 | Published: N/A****MEDIUM SEVERITY**

Directory traversal vulnerability in strong.exe program in NAI Net Tools PKI server 1.0 before HotFix 3 allows remote attackers to read arbitrary files via a .. (dot dot) attack in an HTTPS request to the enrollment server.

**CVE-2000-0740 — CVSS Score: 5.0 | Published: N/A****MEDIUM SEVERITY**

Buffer overflow in strong.exe program in NAI Net Tools PKI server 1.0 before HotFix 3 allows remote attackers to execute arbitrary commands via a long URL in the HTTPS port.

**CVE-2000-0791 — CVSS Score: 4.6 | Published: N/A****MEDIUM SEVERITY**

Trustix installs the httpd program for Apache-SSL with world-writeable permissions, which allows local users to replace it with a Trojan horse.

## 4. MITRE ATT&CK; Mappings

| Port | Technique ID | Technique Name            | Tactic    |
|------|--------------|---------------------------|-----------|
| 53   | T1046        | Network Service Discovery | Discovery |

|     |       |                                   |                |
|-----|-------|-----------------------------------|----------------|
| 443 | T1190 | Exploit Public-Facing Application | Initial Access |
|-----|-------|-----------------------------------|----------------|

## 5. Immediate Remediation & Hardening

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- **Patch & Update Software:** Immediately apply vendor security updates for all identified outdated services and libraries.
- **Enforce Strict Firewall Policies:** Restrict inbound network access to essential ports only; block unused open ports.
- **Implement Strong Authentication:** Enforce multi-factor authentication (MFA) and disable password-based SSH authentication in favor of key pairs.
- **Continuous Monitoring & Auditing:** Enable continuous log monitoring, intrusion detection systems (IDS), and regular automated vulnerability assessment.